

Cardiac Real-Time Imaging Using SENSE

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Sensitivity encoding is used to improve the performance of real-time MRI. The encoding efficiency of single-shot and segmented echo-planar imaging is tripled by means of a 6-element receiver coil array. The feasibility of this approach is verified for double oblique cardiac real-time imaging of human subjects at rest as well as under physiological stress. Sample images are presented with scan times per image down to 13 msec at a spatial resolution of 4.1 mm, and 27 msec at a resolution of 2.6 mm. Moreover, multiple slice real-time imaging is demonstrated at a rate of 38 double-frames per second. Magn Reson Med 43:177–184, 2000. © 2000 Wiley-Liss, Inc.

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In recent years, substantial efforts have been made to increase temporal resolution in magnetic resonance imaging. In this process, the notion of real-time imaging has emerged, denoting the continuous acquisition of entire images at a rate sufficiently high to resolve dynamic processes, such as the motion of tissue or interventional devices. Images acquired in this fashion have also been designated instant (1) or snapshot images (2).

The real-time mode is of particular interest in cardiac imaging, with a whole range of intriguing applications. Continuous heart imaging with high frame-rates could give unprecedented insight into cardiac dynamics, both for diagnostic purposes and basic research. In addition, the real-time approach has the important practical advantage that, unlike conventional cardiac imaging, it does not require ECG triggering, breath-holding, or navigator gating. In fact, a real-time examination can in principle be conducted without external devices or patient cooperation.

The temporal resolution required to resolve cardiac motion depends not only on tissue velocity, but also on the spatial resolution desired. A straightforward estimation yields that tissue must not travel more than roughly the size of a voxel during the acquisition of one frame. Otherwise blurring occurs, reducing the actual spatial resolution. Therefore, to reasonably improve spatial resolution, the frame-rate must be increased along with the size of the acquisition matrix. More specifically, unlike in conventional MRI, the k -space area covered per unit time must grow as the inverse cube of pixel size. This observation illustrates why the real-time concept has formed a tremendous challenge to gradient performance and sequence design.

Cardiac real-time imaging began as early as in 1987, when Rzedizian et al. (1) and Chapman et al. (3) used

single-shot echo-planar imaging (EPI) to achieve spatial resolutions of about 4 mm at scan times per image down to 40 msec. Better overall image quality was accomplished by McKinnon (2.5 mm/110 msec) (4), introducing segmented EPI. Later reports include those of Davis et al. (3.7 mm/50 msec) (5), Kerr et al. (2.5 mm/140 msec) (6), and Stuber et al. (2.5 mm/75 msec) (7). Increases in gradient strength and, more importantly, in gradient switching rates have gradually improved the performance of real-time imaging. Further significant enhancement of gradient power, however, is to be doubted in view of the problem of nerve stimulation by gradient switching. Therefore, the time efficiency of collecting data by mere gradient encoding seems to be approaching a fundamental limitation.

Recently, a complementary fashion for encoding resonance signal, namely by spatially varying receiver sensitivity, has received increasing attention (8,9,10). The SENSE scheme (10) increases the time efficiency of encoding by using an array of parallel receiver coils. In this work we report the use of sensitivity encoding for the purpose of enhancing real-time cardiac imaging. With a 6-element coil array, reductions of scan time to less than one-third have been achieved with respect to mere gradient encoding. The resulting increase in frame-rate has enabled real-time imaging of cardiac stress as well as simultaneous real-time imaging of two cardiac sections.

MATERIALS AND METHODS

Sensitivity Encoding

In this section a brief summary of the SENSE technique for the case of Cartesian sampling of k -space is given. For a complete and detailed description we refer to (10).

Sensitivity encoding makes use of spatial information related to the spatially varying sensitivity of receiver coils. Using an array of multiple receivers, this allows reduction of scan time in standard Fourier imaging by increasing the distance between sampling lines in k -space. Doing so, k -space is basically undersampled, resulting in aliasing in conventional single-coil images. That is, each pixel of a single-coil image reflects the sum of signal components from multiple equidistant locations. However, in the aliasing process each component is individually weighted according to local coil sensitivity. Therefore, the distinctness of coil sensitivities to a certain degree enables separation of aliased pixels by means of linear algebra. The unaliased components of an aliased pixel are obtained by appropriate linear combination of the values the pixel exhibits in the different single-coil images:

$$v = (S^H \Psi^{-1} S)^{-1} S^H \Psi^{-1} a,$$

where the aliased and unaliased pixel values are assembled in the vectors a , v , respectively. S is a matrix of

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complex sensitivity values with the entry S_{ij} given by the sensitivity of the i -th coil at the position of the j -th of the superimposed pixels. The matrix superscript H denotes the conjugate transpose, and the matrix Ψ reflects the correlation of noise in the receiver channels and is assessed separately as described in (10). Noise correlation is taken into account for the sake of SNR optimization. For easier implementation Ψ may as well be set to identity, however at some penalty in final image signal-to-noise ratio (SNR).

The reduction factor R denotes the reduction of the number of phase-encoding steps with respect to full Fourier encoding. R can be an integer or non-integer value between one and the number of coil elements used. The degree of reduction affects the SNR of the final image in two ways. On the one hand, the SNR is bound by the square-root of acquisition time and thus inversely proportional to the square-root of R . On the other hand, in image reconstruction from sensitivity encoded data, noise enhancement occurs in regions where the geometrical relations of coil sensitivities are not ideal. This effect is described by a local geometrical factor g . Thus, at a given location the SNR comparison between a SENSE image and a fully Fourier encoded version is described by

$$SNR^{\text{SENSE}} = \frac{SNR^{\text{full}}}{g\sqrt{R}}.$$

The factor g is a mathematical function of coil sensitivities, of receiver noise correlation, and of the degree of reduction (10), with geometrical factors usually growing as R increases. Clearly, as sensitivity encoding is used here as a means of undoing fold-over, geometry-related noise enhancement also depends on the phase-encoding direction. The choice of favorable phase-encoding directions again crucially depends on the coil arrangement. Generally, for best time efficiency, phase-encoding is performed along the shortest dimension of the desired field-of-view. Consequently, in cardiac imaging the reasonable fold-over directions approximately lie in the transverse plane. Therefore, in cardiac applications the coil arrangement should be optimized such as to favor pixel discrimination in the anterior-posterior and left-right directions. Cranial-caudal aliasing need not be considered, which significantly helps in optimizing the coil arrangement.

Artifact-free SENSE reconstruction relies on accurate knowledge of the individual coil sensitivities. For sensitivity assessment, low-resolution, fully Fourier-encoded reference images are required, obtained with each array element and with a body coil. Element-wise division of the array references by the body coil reference yields raw sensitivity maps. Thus, the homogeneity of the body coil is implicitly made the basis of the homogeneity correction carried out by SENSE reconstruction. Raw sensitivity maps are refined by a fitting procedure which performs noise elimination and sensitivity extrapolation. Regions that, according to the references, do not contribute significant signal are automatically excluded from SENSE reconstruction. Thereby, it is taken into account that the true degree of aliasing is locally reduced when one or more aliased components are zero. The important consequence

of pixel exclusion is that it reduces g for the remaining pixels. Pixels eligible for exclusion are mainly found outside of the body. However, in cardiac imaging parts of the lung are also frequently identified as not contributing significant signal.

Reference Imaging

In this work, a 64×64 matrix was used for reference imaging. Due to inductive coupling, the body coil and the array of surface coils cannot be operated simultaneously, thus requiring acquisition of the reference images in two steps. In order to avoid the need for breath-holding or triggering, a fast, interleaved approach was taken. Half-Fourier gradient EPI with three segments was used, requiring six echo-trains overall, which were alternately read out with the array and the body coil. With this strategy the data for one complete set of reference images were collected in 74 msec (13 echoes per segment, 2×3 segments, $TE = 3.6$ msec, $\alpha = 20^\circ$). For each slice of interest, 20 such sets of references were collected successively. Although, in principle, a single set enables the generation of sensitivity maps, it is advantageous to average sensitivity information from a range of heart phases. On the one hand this improves SNR, making raw sensitivity values more reliable. On the other hand, sensitivity can only be assessed in regions where signal is present in the reference images. In this regard it is beneficial to consider a range of relative positions and extensions of the myocardium for sensitivity determination. In the construction of sensitivity maps, weighted averaging of raw maps from different heart phases was done after tissue contrast had been removed by division by the body coil references. The reliability of a raw sensitivity value crucially depends on the SNR of the underlying measurements. Therefore, the weighting coefficients were chosen pixel-wise proportional to the modulus of the body coil references.

Real-Time Sequences

Segmented and single-shot gradient EPI sequences (4,11) were used for real-time SENSE imaging. Four protocols with different slice orientations and sequence parameters were created as listed in Table 1. Protocols 1–3 were used for repeated imaging of a single slice. Protocol 4 was designed for multiple-slice real-time imaging by alternate excitation and read-out of two parallel slices with a single echo-train each.

The distance between read-out lines in k -space was increased using SENSE reduction factors $R = 2.7$ – 3.2 . Half-Fourier acquisition was applied to further reduce the number of phase-encoding steps and the effective echo-time. That is, in one half of k -space only about the central quarter was sampled, resulting in overall k -space coverage of 58–65%. Finally, for the sake of uniform EPI segmentation, a varying number of high- k_y profiles was skipped and zero-padded. The resulting loss in resolution in the phase-encoding direction was taken into account in the calculations for Table 1. Parameters were generally chosen so as to yield nearly isotropic in-plane resolution.

In all experiments, 10-mm thick slices were excited at constant flip angle. Flip angles were chosen between 25

Table 1
Protocols for Cardiac Real-Time SENSE

Protocol	1	2	3	4
<i>General</i>				
Anatomic view	short-axis equatorial	short-axis basal	four-chamber	short-axis 2 slices
Image matrix $n_x \times n_y$	128×96	128×96	128×116	64×64
FOV [mm ²]	310.0×232.5	310.0×232.5	390.0×353.4	250.0×250.0
Resolution $\Delta x \times \Delta y$ [mm ²]	2.42×2.72	2.42×2.77	3.05×3.43	3.91×4.26
Scan time per image [ms]	34.5	27.0	31.4	2×13.0
Frame-rate [images/s]	29.0	37.0	31.8	76.9 (38.5)
SENSE reduction factor R	2.67	3.00	3.22	2.67
Flip angle α	30°	30°	25°	35°
Figure	2, 3	4	5	6
<i>Reconstruction</i>				
# k_y acquired m_{acq}	21	18	22	14
# k_y after HF completion m_{HF}	32	28	32	22
# k_y after zero-padding m_{zp}	36	32	36	24
# y after DFT n_{alias}	36	32	36	24
# y after SENSE reconstr. n_y	96	96	116	64
<i>EPI sequence</i>				
# EPI segments	3	2	2	1
# EPI echoes per segment	7	9	11	14
Effective TR [ms]	11.5	13.5	15.7	13.0
Readout window per TR [ms]	7.1	9.1	11.2	8.9
Effective echo time TE [ms]	3.7	3.7	4.8	3.5
<i>Noise</i>				
Maximum g	2.0	2.5	2.0	1.6/1.9
Mean g	1.3	1.6	1.3	1.2/1.3

and 35 degrees so as to create optimum in-flow contrast. The phase-encoding direction was chosen anterior-posterior for the short-axis views and right-left for the four-chamber view, respectively. All sequences included first-order flow compensation.

Materials

Experiments were performed on a 1.5 T Philips Gyroscan ACS-NT15, equipped with Power Trak 6000 gradient amplifiers. The highest slew-rate used was 100 mT/m/msec. The peak gradient strengths amounted to 20 mT/m, with maximum values of 17 mT/m during EPI acquisition. A 6-coil receiver array was used, consisting of two circular ($\varnothing$ 20 cm) and four rectangular (10 cm $\times$ 20 cm) elements. These had been designed and were arranged such as to optimize SNR in typical cardiac views (see Fig. 1), considering both overall sensitivity and geometry-related noise enhancement. To ensure consistency of coil sensitivities, the posterior coils were mounted on the patient support in a flat case, whereas the lateral and anterior coils were fixed to a frame. The frame used was adaptable to patient size and shape, yet rigid against motion related to breathing or exercise, for example. For imaging experiments at increased heart rates, the volunteers were asked to perform physiological exercise inside the tomographer, using a home-built, MR-compatible ergometer.

SENSE reconstruction in real-time is readily feasible using dedicated hardware (12). The images presented in this work, however, were reconstructed off-line on a DEC Alpha workstation.

RESULTS

According to the four protocols listed in Table 1, real-time imaging experiments were performed with healthy volunteers in a free-running mode, i.e., without breath control or ECG triggering. Despite high gradient performance, the examined persons reported no peripheral stimulation.

Figure 2 shows a sample set of consecutive real-time images acquired at rest using Protocol 1 with $R = 2.7$. A short-axis view was chosen at an equatorial level. The images shown were extracted from a series of 200 frames. About one cardiac cycle is covered by 24 images with a

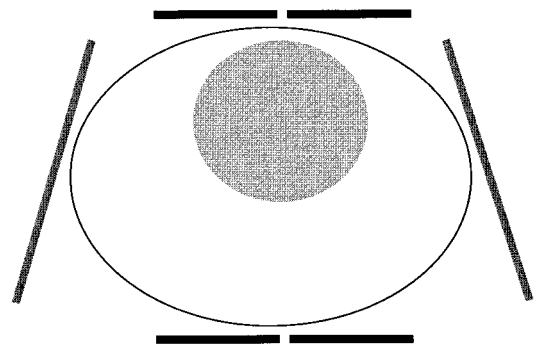


FIG. 1. Configuration of the 6-element receiver array. Two circular elements (20-cm diameter) (gray) were placed laterally. Four rectangular elements (10 cm $\times$ 20 cm) (black) were placed pair-wise on the chest and under the back in a non-overlapping fashion, with their long axes in feet-to-head direction.

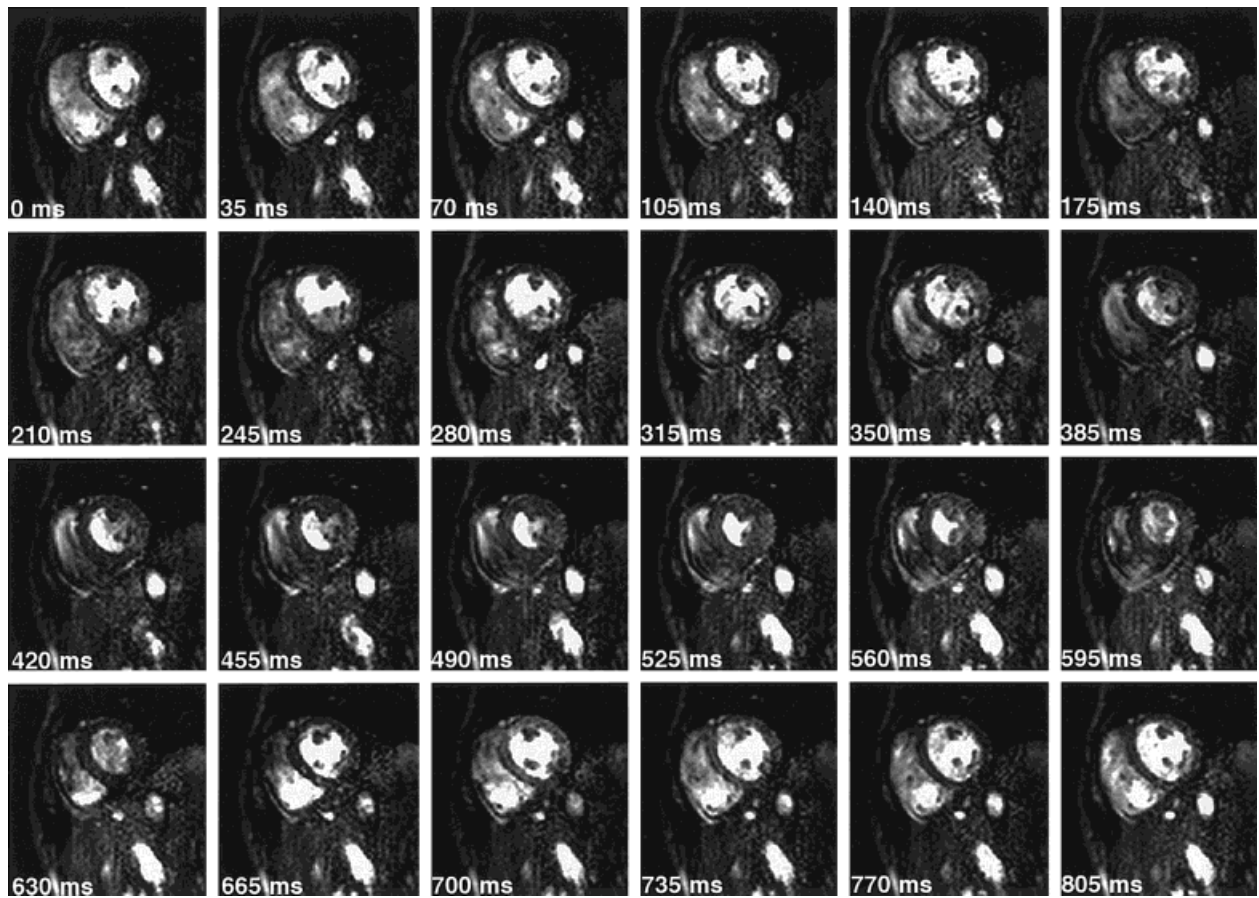


FIG. 2. Twenty-four consecutive real-time SENSE images in a short-axis view, taken from a series of 200 images obtained with protocol 1 (Table 1). Each image was acquired in 35 msec corresponding to a rate of 29 frames/sec. The sequence displayed covers approximately one cardiac cycle. The in-plane resolution is $2.4 \times 2.7 \text{ mm}^2$. Enlarged copies of the two images at 525 msec and 805 msec are shown in Fig. 3.

scan time of 35 msec each. Most of these images exhibit sufficient blood-to-muscle contrast to clearly discern blood and myocardium. The papillary muscles are well depicted. Myocardial dynamics, in particular contraction and heart wall thickening, are readily observed. A surprisingly detailed impression of tissue and blood motion is given by animated display, revealing many details not perceivable in a static representation. The changes in the appearance of the aorta and the vena cava reflect blood pulsation resulting in varying degrees of blood saturation.

Protocol 1 was also used for imaging cardiac stress. Real-time imaging was started at a heart rate of 155 beats/min, immediately following physiological exercise. Figure 3 shows images obtained at stress in two selected heart phases, together with images of the same slice acquired at rest in equivalent phases. Care was taken to choose images obtained at similar diaphragm positions, so as to ensure that the physical slices shown are nearly identical. Despite the increased heart rate and heavy breathing, the quality of the stress images appears just slightly reduced. Figure 3a reveals increased end-systolic contraction and heart wall thickening upon stress, whereas at mid-diastole (Fig. 3b) the myocardium appears a little thinner than at rest. The brighter depiction of the liver in the stress images is due to breathing, which reduces the saturation effect most prominent in static tissue.

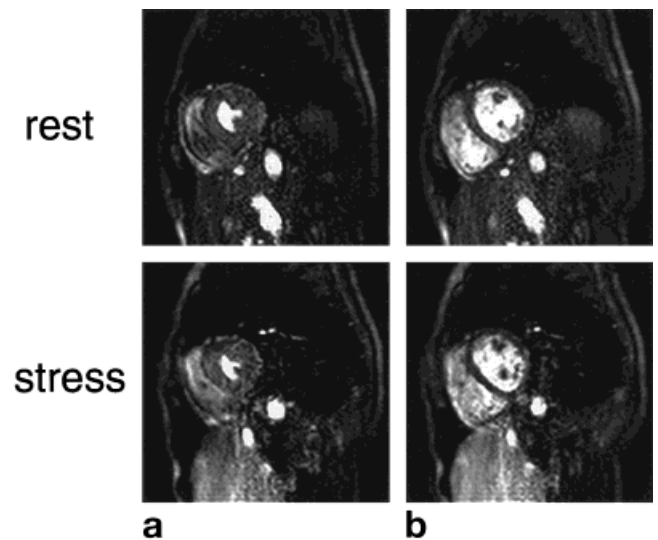


FIG. 3. Real-time images obtained at rest (top) and stress (bottom) in end-systole (a) and mid-diastole (b). The scan time per image was 35 msec at $2.4 \times 2.7 \text{ mm}^2$ in-plane resolution. The heart rates were 70 beats/min at rest and 155 beats/min at stress, respectively. Note that despite the greatly increased heart rate, image quality is only slightly affected by stress. For sequence parameters see protocol 1 in Table 1.

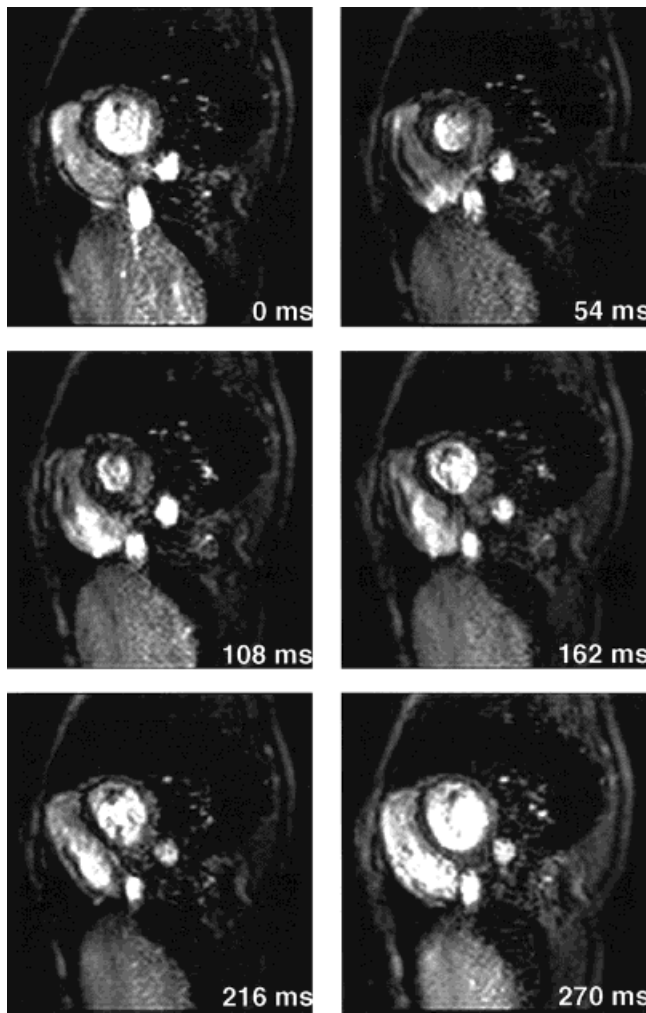


FIG. 4. Real-time images acquired at stress in 27 msec per image (37 frames/sec), showing a short-axis view at a basal level. With respect to Fig. 3, the SENSE factor was increased from 2.7 to 3.0, reducing the scan time per image at preserved spatial resolution. The displayed series covers about one heart cycle at the heart rate of 180 beats/min. Every other image is shown. For sequence parameters see protocol 2 in Table 1.

Preserving the spatial resolution, the scan time per image was reduced to 27 msec by using $R = 3.0$ (protocol 2). The increased frame-rate of this protocol was utilized for imaging exceptional cardiac stress at a heart rate of 180 beats/min in the same volunteer. For this series a short-axis view was chosen at a basal level. Figure 4 shows every other of 12 consecutive frames covering about one heart cycle (324 msec). In these images the contours are less clear than in those obtained with Protocol 1. This is due mainly to the enhanced motion of blood and myocardium, but also related to some reduction in SNR and to the extension of the echo-train. However, the left ventricle still is readily discerned and enables observation of myocardial dynamics.

For imaging a four-chamber view the FOV had to be increased. Therefore, in order to maintain the temporal resolution, higher reduction was applied ($R = 3.2$) and the spatial resolution was slightly reduced (Protocol 3). Figure 5 shows images of three selected heart phases acquired in 31 msec each. The closed bicuspid (black arrow) and tricuspid (white arrow) valves are visible immediately prior to rapid filling in diastole. Rapid inflow from the atria into the ventricles is observed in Fig. 5b), with bright pixels depicting blood not saturated by multiple previous excitations. The late-diastolic sample (Fig. 5c) shows maximum filling of the ventricles. The septum is well discerned in all images, whereas the left ventricular wall is not as distinct. In cine display of the four-chamber series, the varying pre-saturation of blood provides insight into the directions and changing velocities of blood flow.

Using a smaller matrix size of 64×64 , the number of echoes required per image becomes small enough to reasonably perform single-shot EPI in the heart. With Protocol 4 the collection of data for an image was completed within 13 msec, yielding 77 images per second at a resolution of $3.9 \times 4.3 \text{ mm}^2$. Using this mode, two parallel short-axis views at a distance of 30 mm were alternately imaged at a rate of 38 double-frames per second. Figure 6 shows two double-frames out of the multiple-slice real-time series, corresponding to two selected heart-phases. According to these samples, imaging still worked robustly at this rate, depicting the left ventricle with adequate contrast and free of specific artifacts. Note that each of these images is reconstructed from only 14 echoes actually acquired.

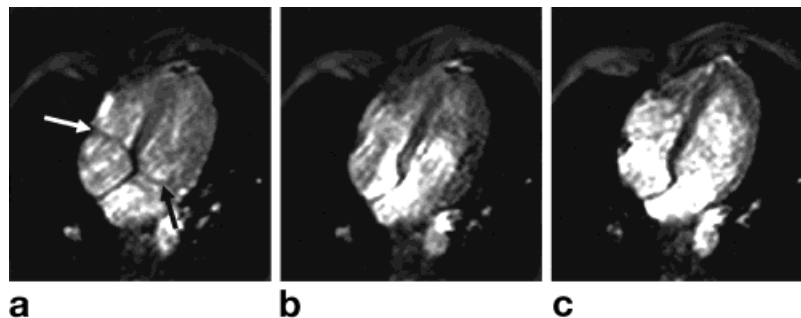


FIG. 5. Three samples from a real-time series of a four-chamber view. The scan time per image is 31 msec (32 frames/sec), the in-plane resolution is $3.0 \times 3.4 \text{ mm}^2$ (see protocol 3 in Table 1). The selected images correspond to heart phases immediately before (a), during (b), and after (c) the rapid filling phase. Note the closed bicuspid (black arrow) and tricuspid valves (white arrow) visible in a. The bright spot in the right ventricular wall in a is not related to SENSE reconstruction. It is also present in the aliased images and follows myocardial motion. Located near respiratory tracts, the spot most probably is a susceptibility artifact typical for EPI imaging.

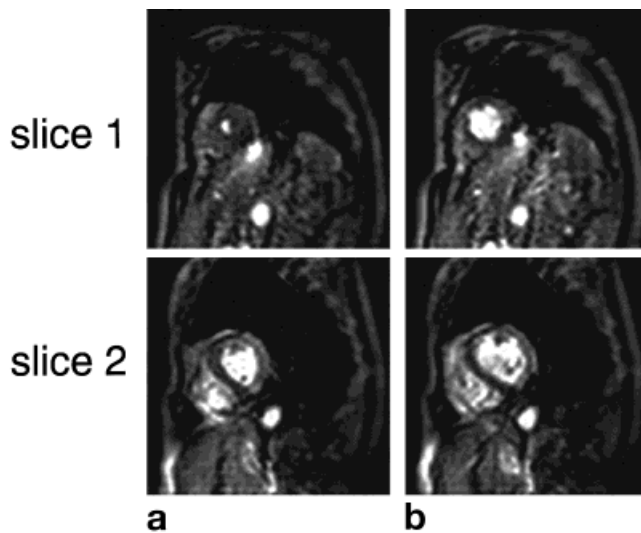


FIG. 6. Simultaneous real-time imaging of parallel slices in short-axis orientation. An apical (1) and an equatorial (2) slice were imaged alternately at a rate of 77 frames/sec (13 msec per image), corresponding to 38 double-frames/sec. Two selected heart phases (a,b) are shown. For sequence parameters see protocol 4 in Table 1.

In order to demonstrate the particular behavior of the SNR in SENSE imaging, the geometrical noise factor g was calculated for the setups used in this work. Figure 7 shows the resulting g -maps for Protocols 1, 2, and 3. The underlying slice orientations and FOVs are illustrated by corresponding reference images displayed in the top row of the figure. These images are intensity corrected samples from the low-resolution reference series acquired for sensitivity calibration. The representations of g reveal replication patterns related to the aliasing effect. Geometry-related noise enhancement is particularly low where aliasing involves regions which do not contribute signal. In the g -maps the beneficial effect of voxel exclusion is prominent in lung regions, and in parts of the FOV where the chest diameter is reduced. Note the correspondence between peaks in

g -values and visible noise enhancement in SENSE images, comparing, e.g., Figs. 7a and 3a.

For quantitative analysis, a rectangular region-of-interest (ROI) was selected in each FOV to focus on the heart alone. For the first three protocols, these ROIs are depicted in Fig. 7. The maximum and mean values of g within the ROIs are listed in Table 1. The maximum values range up to 2.5, whereas the mean values are between one and two. Comparing g as assessed for Protocols 1 and 3, it should be noted that the results are very similar, despite the larger reduction factor in the long-axis view. This demonstrates the influence of different phase-encoding directions, as well as the significance of the relations between FOV and coil size.

DISCUSSION

The results presented in this work demonstrate the capability of SENSE to speed-up cardiac real-time imaging. In this section image quality and some methodological aspects are discussed.

In the protocols discussed here, the blood-to-muscle contrast is based mainly on unequal degrees of saturation of blood and heart tissue by previous excitations. This mechanism has been found to work well in most of the real-time images presented. In some images the myocardium exhibits slightly enhanced signal related to bulk through-plane motion. Most variations in contrast, however, are to be attributed to the complex patterns of blood motion, namely to varying degrees of through-plane flow, reverse flow, and turbulence. In short-axis views the distribution of blood signal intensity is more homogeneous than in four-chamber views. In the latter blood in the atria and the equatorial parts of the ventricles tends to appear brighter than near the apex. This immediately reflects the path which blood travels after entering the atria in an unsaturated state. The generally inhomogeneous appearance of blood may be considered merely an inconvenience in the interpretation of static images. However, as intensity patterns reflect local coherence of blood flow, animated

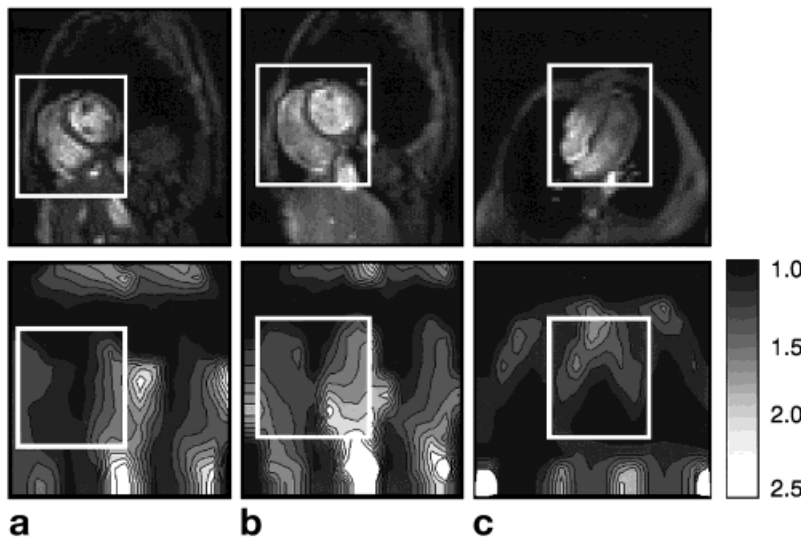


FIG. 7. Geometrical noise enhancement in SENSE. Column a, b, and c correspond to Protocols 1, 2, and 3, as described in Table 1, respectively. The top row illustrates slice orientations and FOVs, showing low resolution reference images. The bottom row displays corresponding maps of the noise enhancement factor g , using the gray-scale given on the right. ROIs were added to evaluate the geometrical factor in the actual heart region. The maximum and mean values of g in these ROIs are listed in Table 1.

display gives a lively impression of flow dynamics. The frame-rates achieved with SENSE enable smooth animation even in slow-motion, revealing subtle details of heart wall motion.

As suggested in the introduction, spatial and temporal resolution in real-time imaging ought to be matched depending on the peak velocities of the structures to be examined. A straightforward requirement is that tissue should not travel more than the size of a pixel during the acquisition of data for one entire image. According to this criterion the single-slice experiments performed with Protocols 1–3 are appropriate for dealing with peak velocities of 8–10 cm/s. This is a reasonable range for resolving myocardial dynamics (13) and breathing motion, as well as moderate in-plane flow. Similarly, Protocol 4, designed to cover two slices, is suited to resolve structures moving at velocities up to 30 cm/s. Observations of fast blood flow, however, are hampered in protocol 4 by the coarse spatial resolution which is insufficient to depict the mostly tiny signal patterns created by transient flow coherence. From the aforementioned principle it is clear that, if myocardial motion is to be visualized, the spatial resolution cannot be improved much beyond 2.5 mm unless acquisition efficiency is further increased. This may be achieved by using stronger gradients or by potential increases in R .

We have demonstrated reductions of scan time by factors between 2.7 and 3.2 with respect to mere Fourier encoding. For each protocol the SENSE reduction factor was chosen by trade-off between frame-rate, spatial resolution, and SNR. In doing so, it has been found that careful tuning considerably benefits from the possibility of non-integer reduction. Generally, SENSE reconstruction works at arbitrary reduction factors between one and the number of coils used. However, beyond the factors reported one faces critical drawbacks in SNR mainly due to the increasing geometry-related noise penalty. In this work, only short-axis and four-chamber views are demonstrated. In transverse, coronal, and sagittal imaging as well as in two-chamber views, the behavior of g is similar as long as the phase encoding direction is chosen properly. Therefore, with the 6-coil array used, reduction factors of about three are readily accomplished in all orientations common to cardiac imaging.

It remains to further study to which extent reduction factors can reasonably be increased by using more receivers. In such investigations it is important to also stick to the conventional means of minimizing scan time. Otherwise, the nominal reduction factor will exaggerate the savings in scan time actually enabled by simultaneous signal acquisition. In particular, the phase-encoding direction must be carefully chosen. Conventional imaging with a rectangular FOV is fastest with phase-encoding along the shorter dimension of the FOV. Sensitivity encoding, on the other hand, is easier along the longer dimension. This is because the coil elements must have mutually distinct sensitivity profiles and at the same time be large enough to exhibit sufficient sensitivity at object depth. These goals can better be combined by arranging coils along the longer dimension. However, while phase-encoding in this direction may enable higher nominal reduction, it is usually not a reasonable choice in terms of minimizing effective scan time.

It has been demonstrated that reference imaging for sensitivity determination can be performed without ECG triggering and breath control. Thus, in principle a real-time examination using SENSE can be conducted entirely without external devices or patient cooperation. One option to further reduce the duration of reference acquisition is to dispense with the body coil image. Correct maps of relative sensitivity are then obtained by using the root-of-sum-of-squares of the array references instead. A drawback is that the intensity calibration implicitly carried out by SENSE reconstruction then reflects the net sensitivity of the array. However, in this case the standard methods for intensity correction are still applicable.

Individual slice-wise reference scans may become inconvenient in successive real-time SENSE imaging of multiple slices, in particular when the slice orientation is chosen interactively. For this purpose 3D sensitivity maps created in one initial effort are more practical. These can still be based on non-triggered reference scans of the kind used in this work for 2D maps.

A subtle, yet important, result of this work is that half-Fourier EPI was successfully used in combination with SENSE. For image reconstruction from sensitivity encoded data it is crucial to consider coil sensitivities in their complex form. In particular, in the processing of reference as well as aliased data, the local phase of coil sensitivity must endure the otherwise essential EPI phase correction. Therefore, phase correction was based solely on the phase differences between even and odd read-outs according to pre-shots without phase-encoding. No absolute phase correction was performed, thus preserving coil-related phases. The half-Fourier principle is critical in that it is based on the assumption of complex conjugate symmetry of k -space, implying the absence of any continuous phase variation in the spatial domain. Typically, the center of k -space is sampled symmetrically in half-Fourier techniques to allow for moderate phase drifts. The results presented in this report suggest that this concept is applicable to the phase variations typical for surface coils. The latter apparently are preserved with sufficient fidelity, even when aliasing occurs.

Scan time reduction in cardiac imaging by means of multiple receivers has also been reported using the SMASH technique (14). SMASH uses Cartesian sampling with integer reduction and differs from the corresponding SENSE procedure in that it operates in k -space rather than in image domain. The basic idea of SMASH reconstruction is to approximate low-order harmonics of the FOV by linear combination of coil sensitivities. As a consequence, the method imposes different criteria for coil design and positioning. The practical differences between the two techniques have been discussed in more detail in (10).

CONCLUSION

According to the results of this work, SENSE substantially enhances the performance of real-time imaging. The resolutions thus accomplished give improved insight into myocardial motion and even a glimpse of cardiac blood dynamics.

The actual use of real-time images of this kind is yet to be assessed in physiological and clinical studies. One in-

triguing application is real-time monitoring of diagnostic stress tests using dedicated reconstruction hardware and on-line display (12). It has been demonstrated that image quality as obtained with the described protocols is little affected by stress conditions. Early detection of stress-related ischemia could be enabled by monitoring several cardiac views, either switching between slices with some dwell interval, or using the multiple-slice mode discussed in this report.

In this work the sensitivity encoding concept has enabled reduction of the scan duration per image to about one-third with respect to mere gradient encoding. This improvement is achieved independently of gradient performance. Therefore, in view of physiological limits to k -space speed, the approach provides a promising complement and alternative to further enhancement of gradient systems.

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